Lesson 2.6

Proving Angle Relationships

Four facts you can use as reasons

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Right Angles Congruence Theorem: all right angles are congruent.

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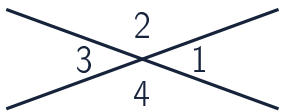
Vertical Angles Congruence Theorem: vertical angles are congruent.

Right Angles Congruence Theorem: all right angles are congruent.

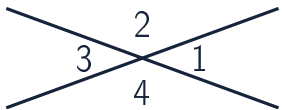
Congruent Supplements / Complements: two angles supplementary (or complementary) to the **same** angle are congruent.

Why vertical angles are congruent

$m\angle 1 + m\angle 2 = 180°$ Linear Pair Postulate



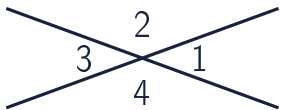
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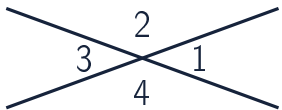


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$\angle 1 \cong \angle 3$ Congruent Supplements Theorem

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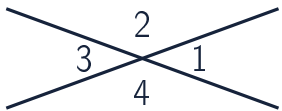
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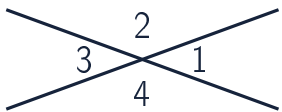
So vertical angles are congruent — that's the whole proof.

Use it to find angles

$m\angle 1 = 3x + 10, \quad m\angle 3 = 5x - 30$



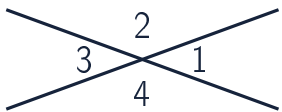
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Vertical angles are congruent:
$3x + 10 = 5x - 30 \Rightarrow 40 = 2x \Rightarrow$
$x = 20$

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$m\angle 1 = 3(20) + 10 = 70^\circ$,
$m\angle 2 = 180 - 70 = 110^\circ$

Watch out

Watch out! Vertical angles are **across** from each other — they are equal. Angles **next to** each other on a line are a linear pair — they add to 180°, they are not equal.

Your turn! Pause the video and try it.

$\angle A$ and $\angle B$ are both supplementary to $\angle C$.
$m\angle A = 2x + 15$, $m\angle B = 3x - 5$. Find x and $m\angle A$.

Your turn! Pause the video and try it.

$\angle A$ and $\angle B$ are both supplementary to $\angle C$.
$m\angle A = 2x + 15$, $m\angle B = 3x - 5$. Find x and $m\angle A$.

Congruent Supplements: $2x + 15 = 3x - 5 \Rightarrow x = 20$,
$m\angle A = 55°$

Remember

- Linear pair: adds to 180°. Vertical angles: congruent.
- All right angles are congruent.
- Supplementary (or complementary) to the **same** angle $\Rightarrow$ congruent.

Now practice at your level:

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